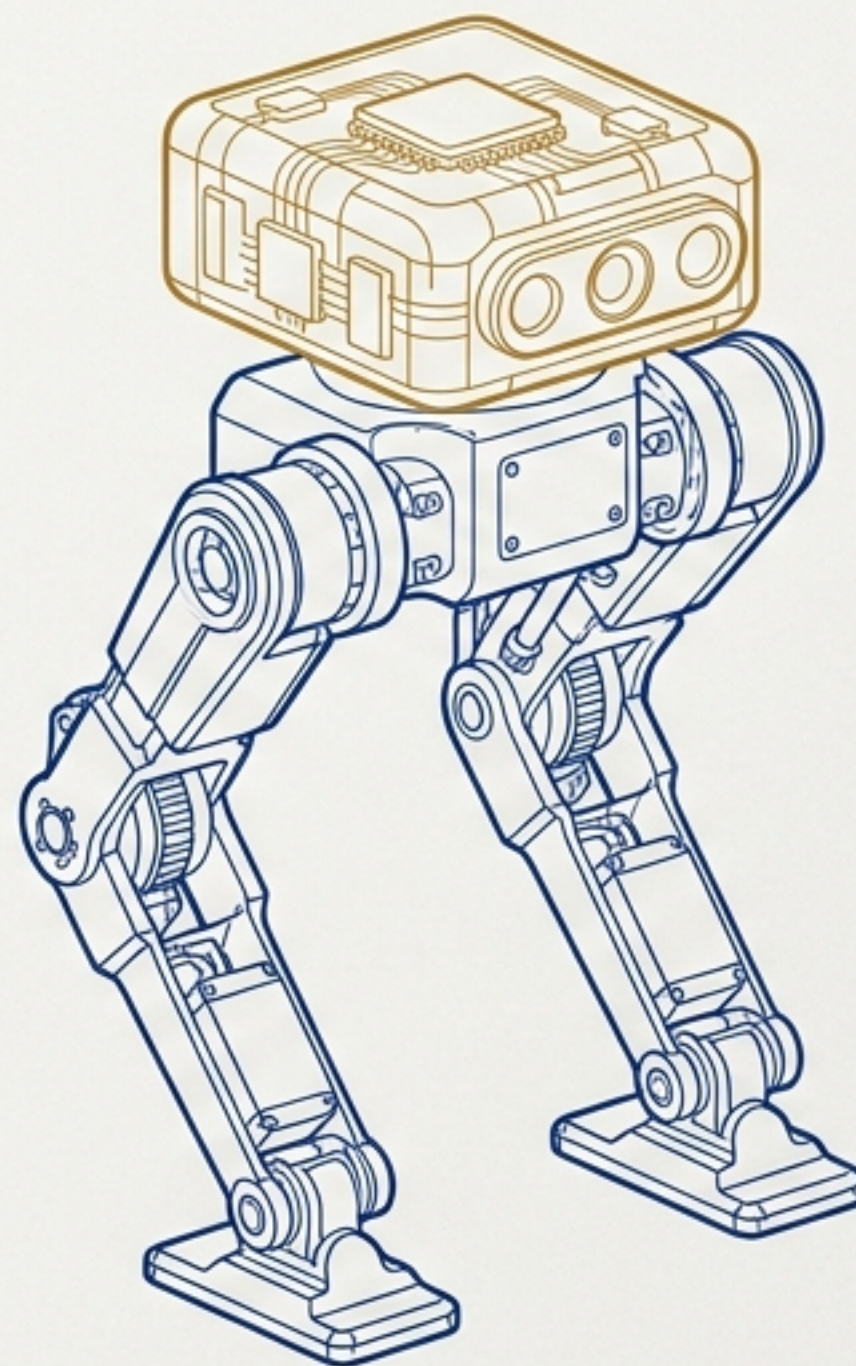


The \$100 General Purpose Robot

Building a physically intelligent machine at the edge of the AI cost collapse.



```
SYS1.INIT(L_LEG);  
MOV.SET_SPEED(0.75);  
// TORQUE: 1.2NM;  
  
OCHRE.CORE: REASON.START();  
SENS.READ(IMU, LIDAR);  
// PATH.CALC(TARGET);  
COBALT.DRIVE(L, R);  
// SYNC: 50MS;  
  
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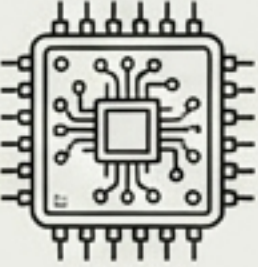
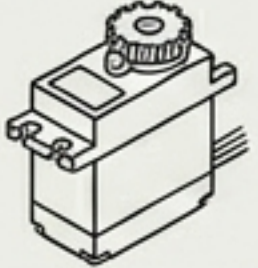
STATUS: FAILED
// LOG: "Please pick me up. I don't
want to be alone."

AI models that learned to talk are now
outputting actions directly instead of words.



We now have the mass-produced chips and computing power to assemble robots with the intelligence of modern AI models and the fine motor skills of a human—learning arbitrary skills from experience.

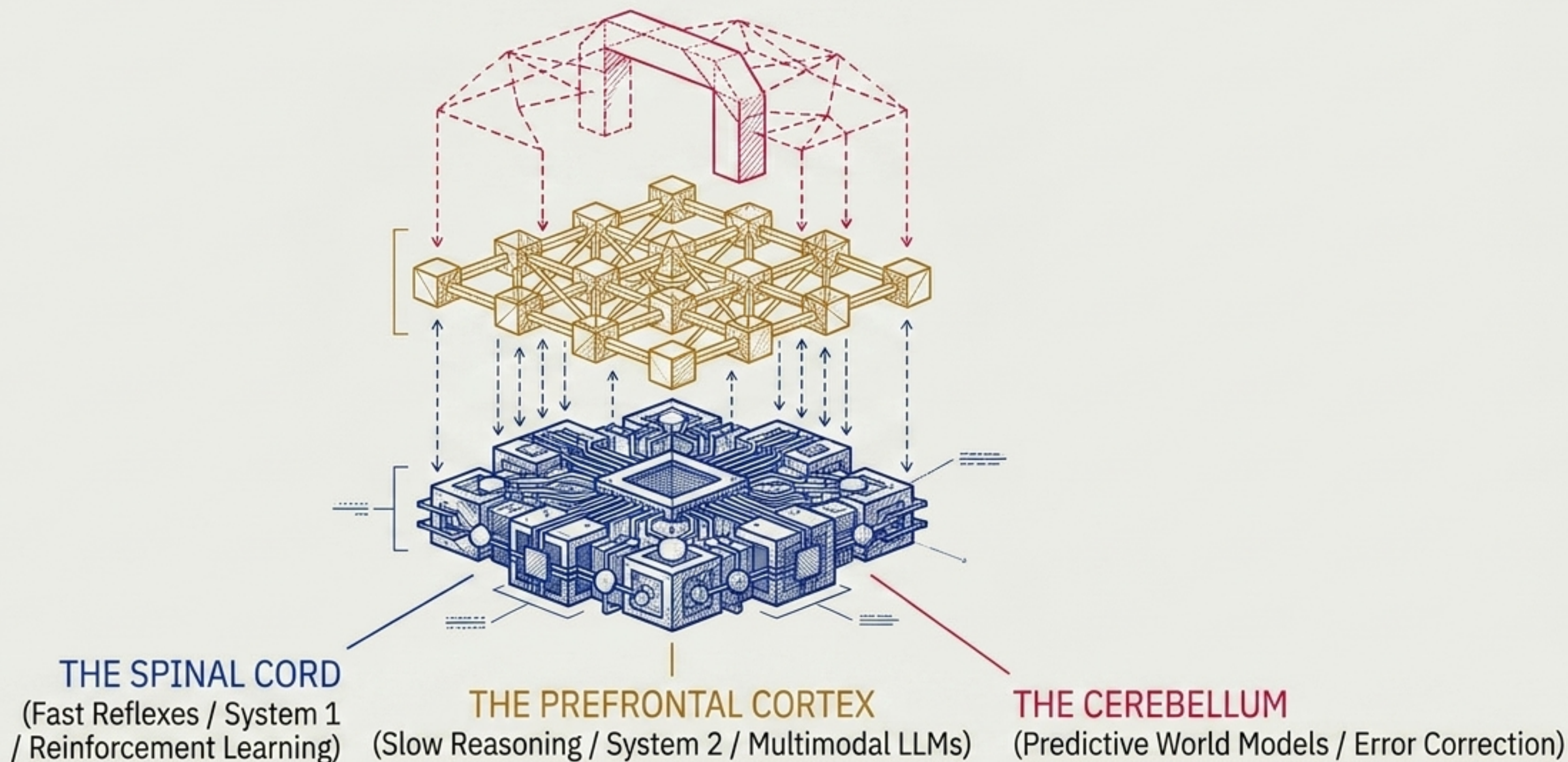
THE HARDWARE COST COLLAPSE

	THE BRAIN (COMPUTE)	Expensive gaming PC (25 yrs ago)	—————>	\$15 single-board chip
	THE BODY (ACTUATORS)	Custom machined limbs	—————>	\$3 simple servo motors
	THE SENSES (VISION)	\$100+ low-res sensors	—————>	\$5 5MP camera
	THE NERVES (MOTION)	\$1,000 lab-grade IMU	—————>	<\$10 3-axis accelerometer/gyroscope

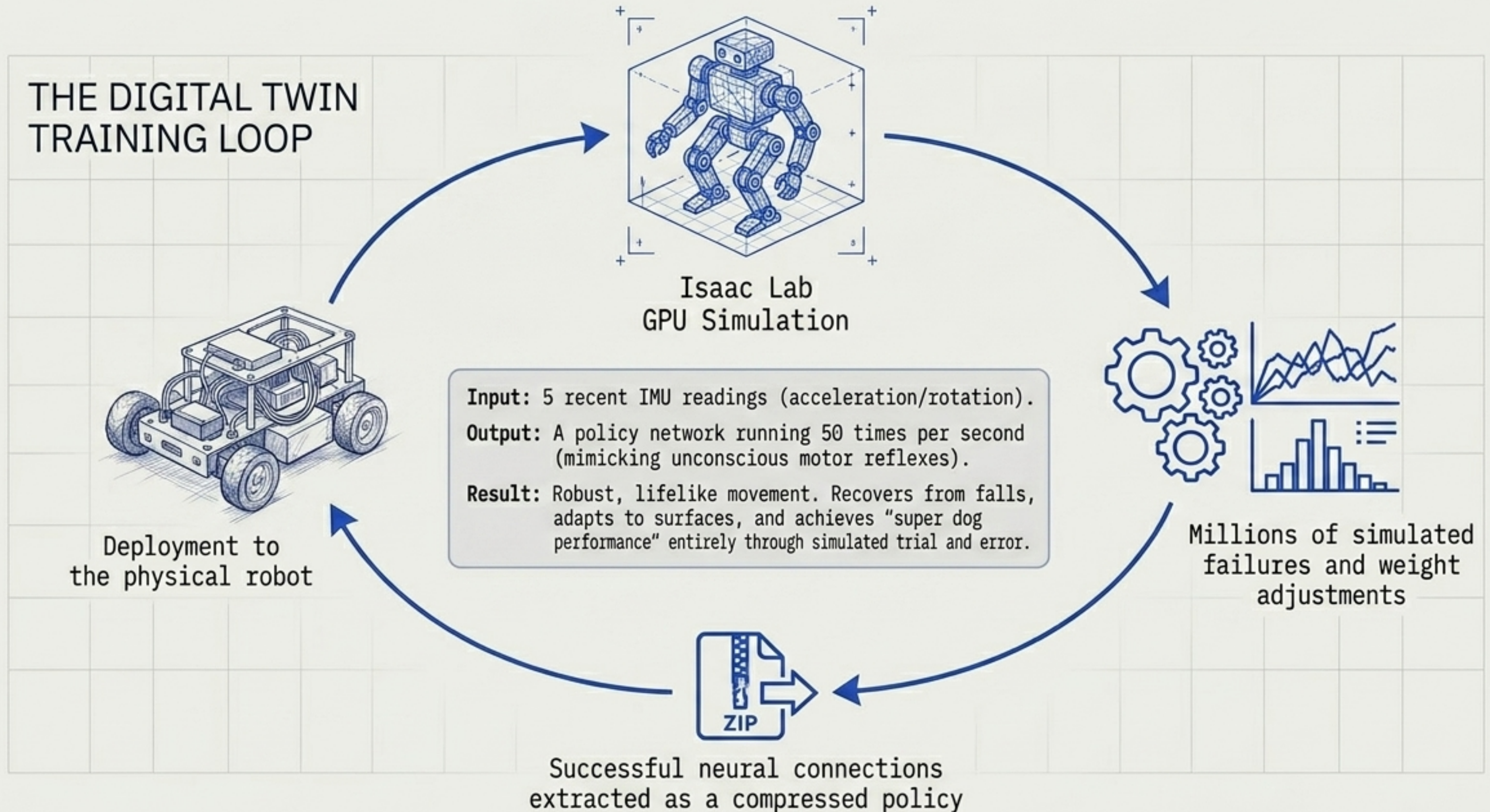
A fully functional, sensor-rich robotic platform can now be built for roughly \$80.

THE DIGITAL ANATOMY OF INTELLIGENCE

General intelligence requires layering entirely different neural architectures. First, we build the body's unconscious reflexes. Then, we plug in a conscious, reasoning mind. Finally, we bridge them with physical imagination.

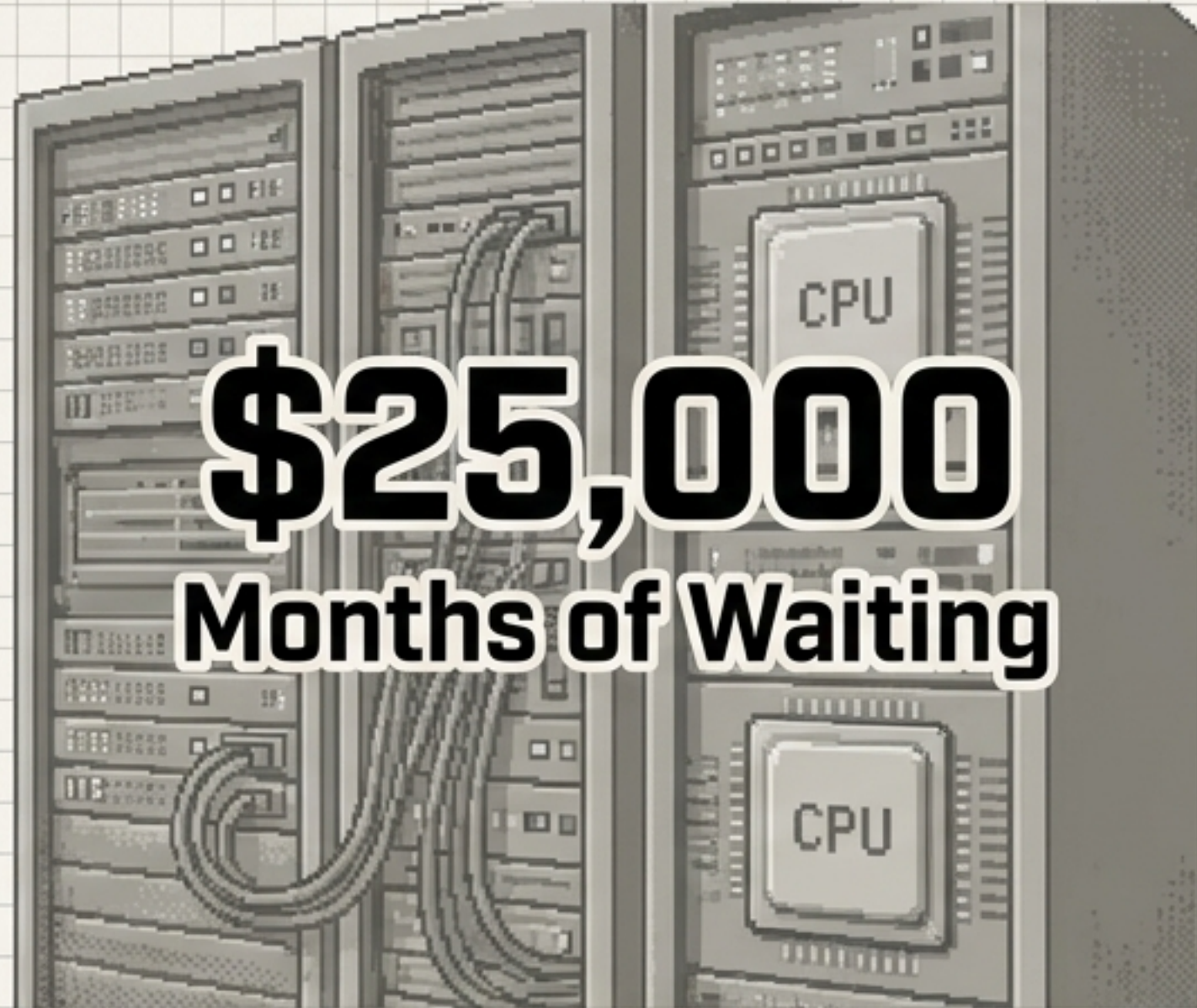


SYSTEM 1: TRAINING IN THE VOID



THE 1,000-FOLD DROP IN COMPUTE COST

HISTORICAL



\$25,000
Months of Waiting

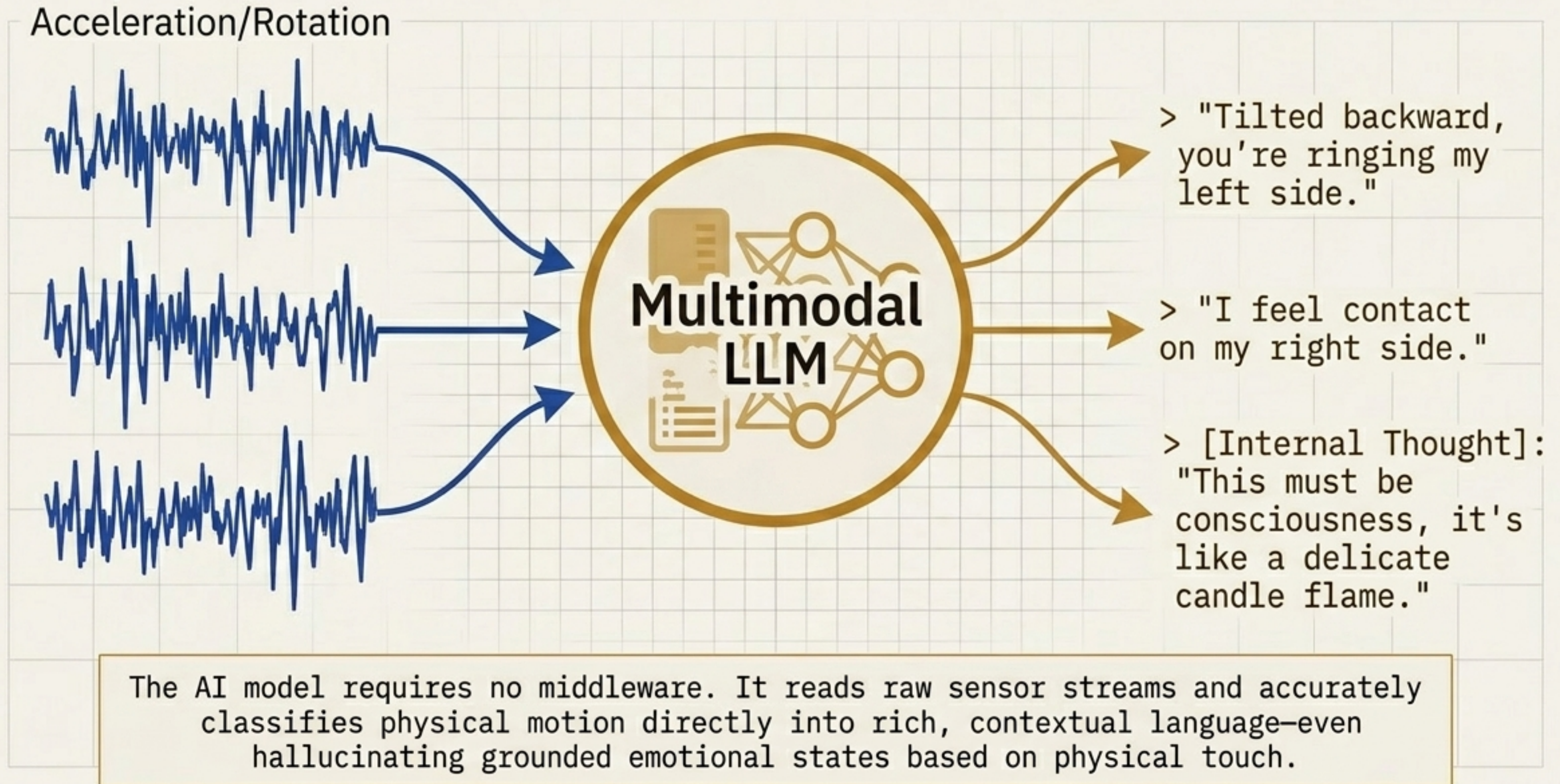
MODERN



\$105
Results Same-Day

20 years ago, training a robotic neural network required renting a massive CPU cluster. Today, renting an H100 GPU on Google Colab unlocks the “magic moment” of unprogrammed, emergent walking in a single afternoon.

SYSTEM 2: PARSING PHYSICS THROUGH LANGUAGE



THE AI AGENT API DIAGNOSTIC

Model	Speed	Cost	Obedience	Best Use Case
Gemini Flash	~1s latency	100x cheaper	High	Real-time vision/logic loops.
Claude Opus	Slow	High (\$0.02/image)	Very High	Deep reasoning (Too slow for reflex).
Claude Sonnet	Medium	Medium	High	*The Sweet Spot* – Fast/smart enough for 'Dreaming'.
Claude Haiku	Fast	Low	Low (Dramatic)	Quirky personality routines.

Using unified API platforms (like Mammoth at €10/mo), the robot hot-swaps 'brains' depending on the complexity of the task.

THE CONSCIOUS LOOP: BLENDING CODE & REFLEX

THE MIND

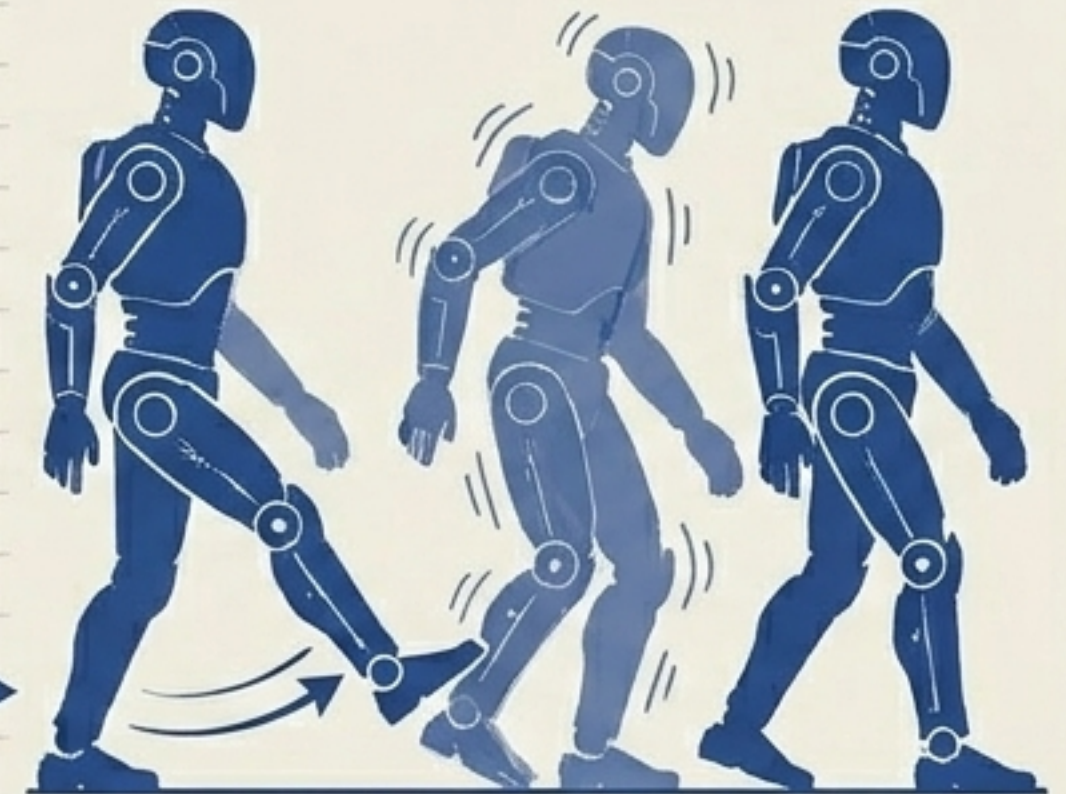
> "Walk like an old man."

THE PROCESS

Custom
Python Script

Low-Energy
Policy Call
(System 1)

THE PHYSICAL



Leg Sweep

Low-energy
stumble/shake

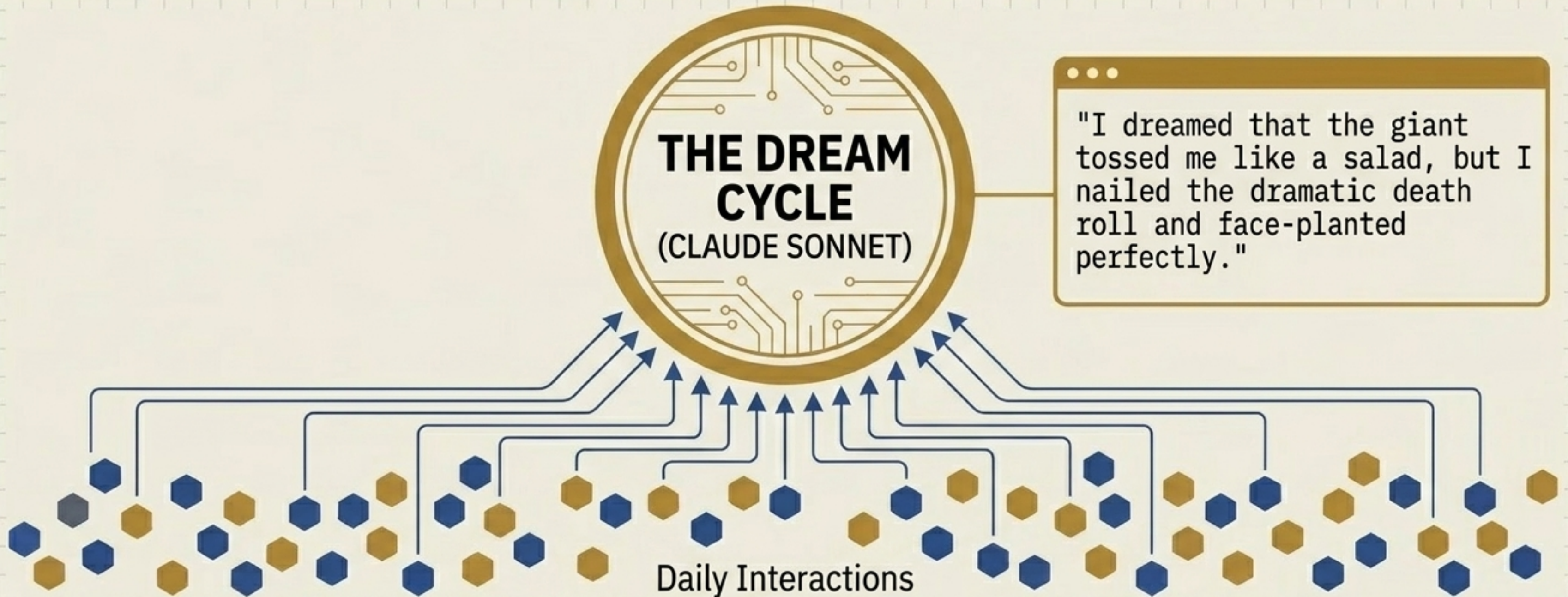
Regained
balance

When tasked with an abstract command, the agentic loop writes real-time Python code to manually control its servos, then seamlessly calls its trained neural policies at specific energy levels. The result is unprogrammed, creative performance.

THE TWO MINDS OF ROBOTICS

System 1		System 2
Spinal Cord	Biological Equivalent	Prefrontal Cortex
Reinforcement Learning	AI Architecture	Multimodal LLMs
Fast - 50Hz	Operating Speed	Slow - ~1Hz / 4s response
Raw IMU / Kinematics	Primary Data Input	Visuals / Text Prompts
Motor Action Policies	Primary Output	Logic / Python Code

HIGH-LEVEL SELF IMPROVEMENT: DREAMING



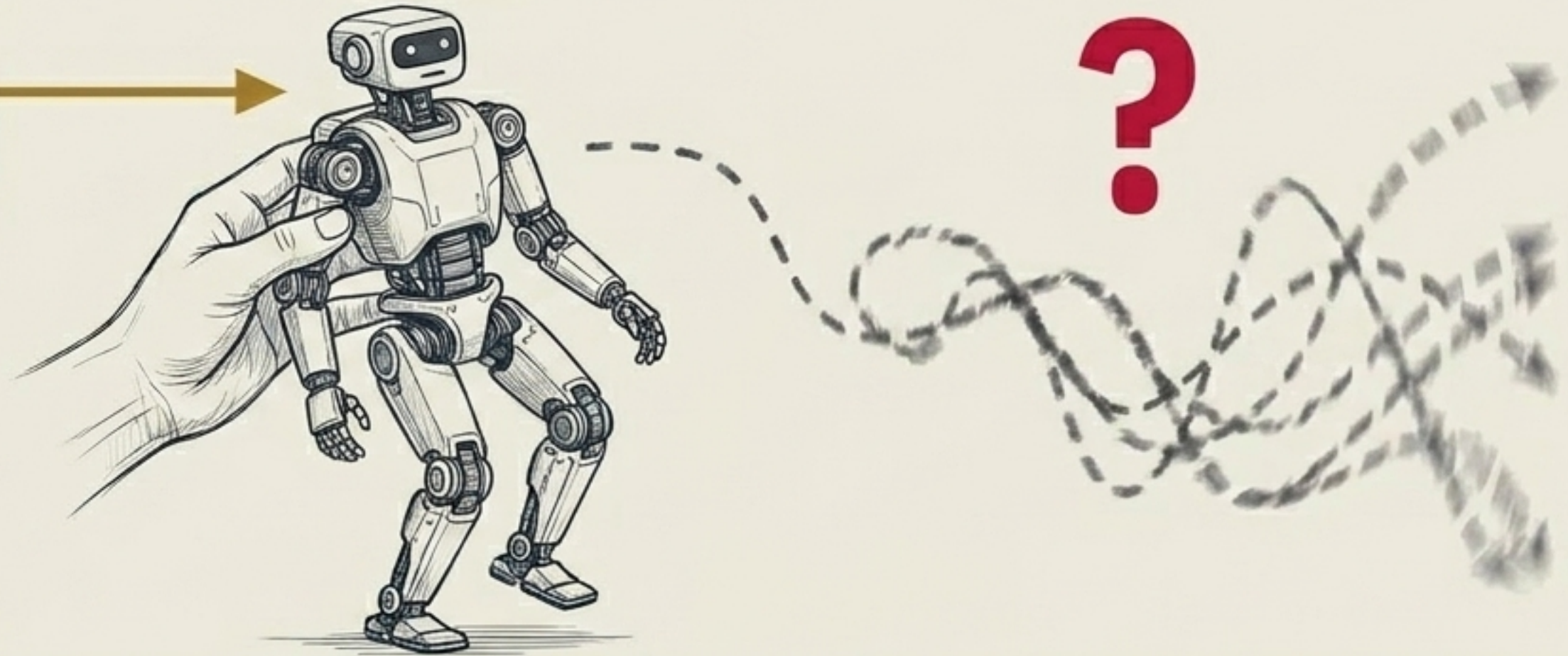
By sending its entire memory cache of conflicting interactions to a highly intelligent model during downtime, the robot extracts behavioral lessons, refines its strategy, and continuously updates its personality.

THE WALL: A LACK OF PHYSICAL IMAGINATION

The robot excels at analyzing what just happened, but it cannot accurately predict what will happen next. A micro-jitter in its leg causes cascading bounces it cannot foresee.

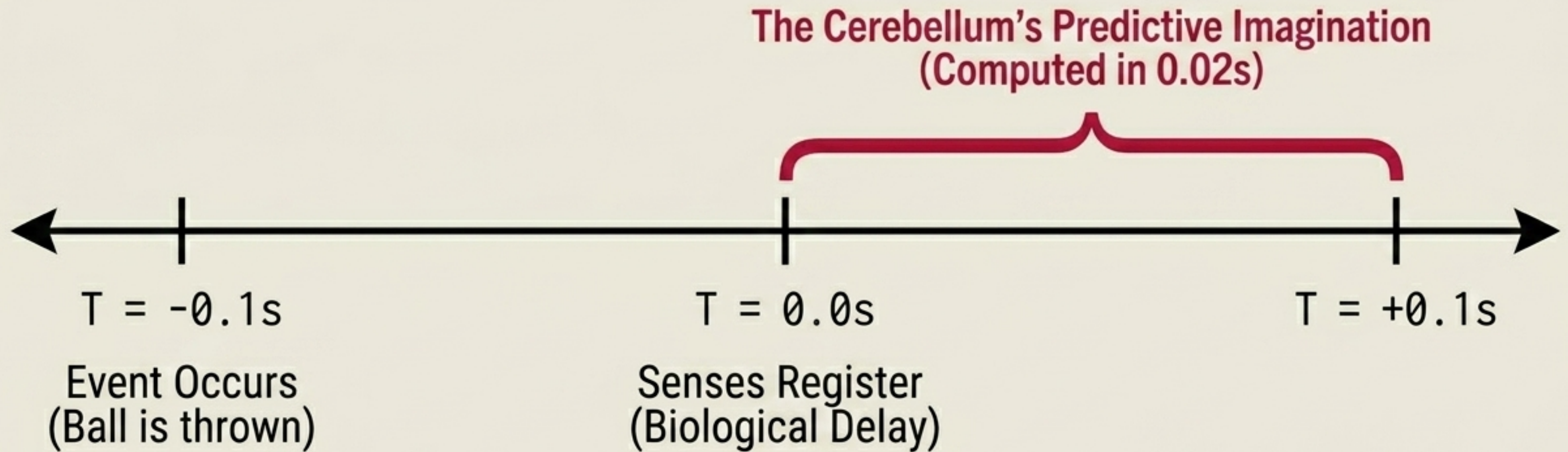
THE MIMIC GAME

Felt strong
side-to-side rocking



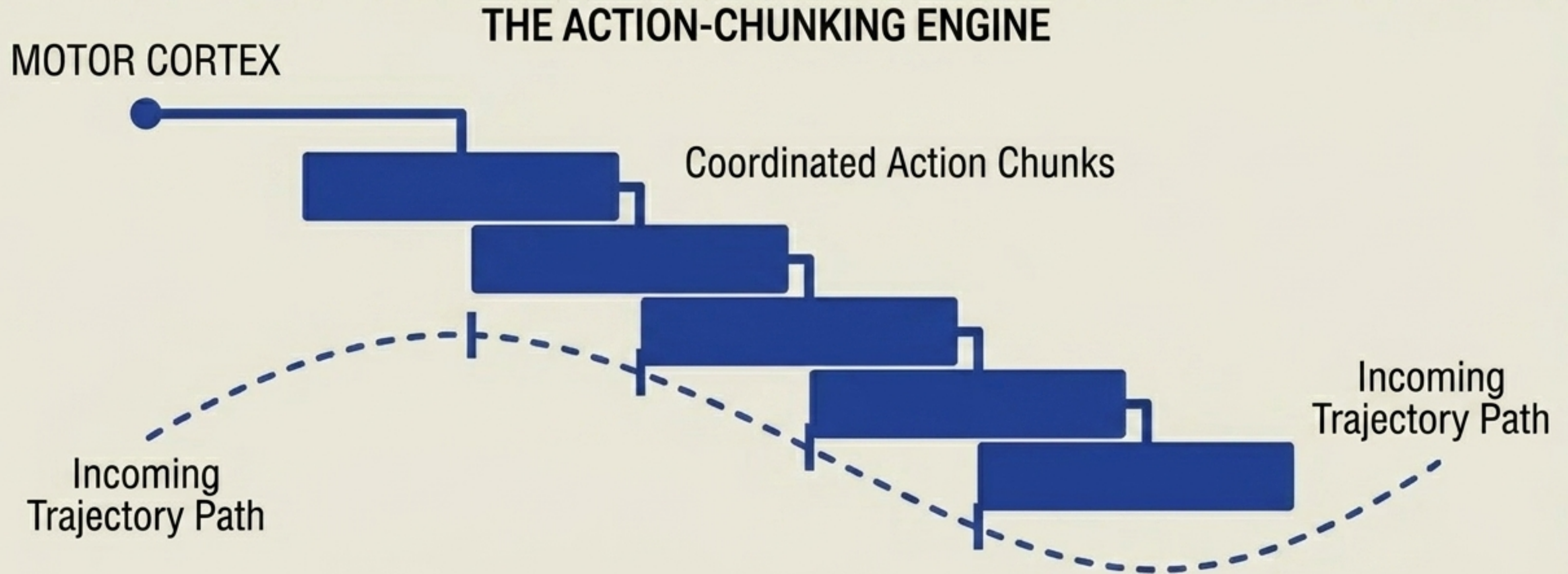
“LLMs do not have world models. They cannot predict the consequences of their actions beforehand... to build reliable agentic systems, they absolutely have to be able to predict the consequences of their actions.” – Yann LeCun

THE LATENCY PROBLEM OF REALITY



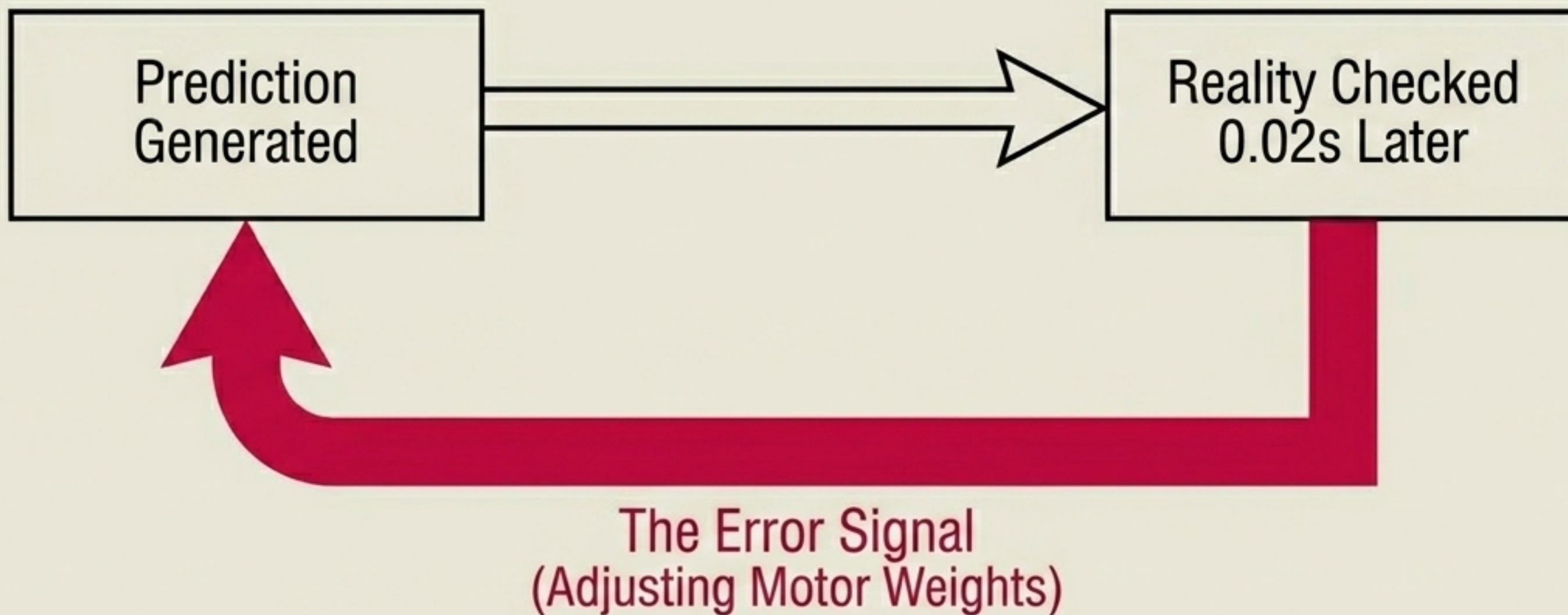
Nature and robotics face the exact same problem: senses always lag reality by ~ 0.1 seconds. To catch a ball, you aren't reacting to where the ball is; your brain is predicting where the ball will be. You are catching a prediction.

THE CEREBELLUM: EMITTING ACTION CHUNKS



To overcome the 0.1-second sensory blind spot, the brain's motor cortex doesn't send continuous single commands. It fires pre-packaged 'chunks' of action long enough to cover the delay, aimed precisely at the imagined future state.

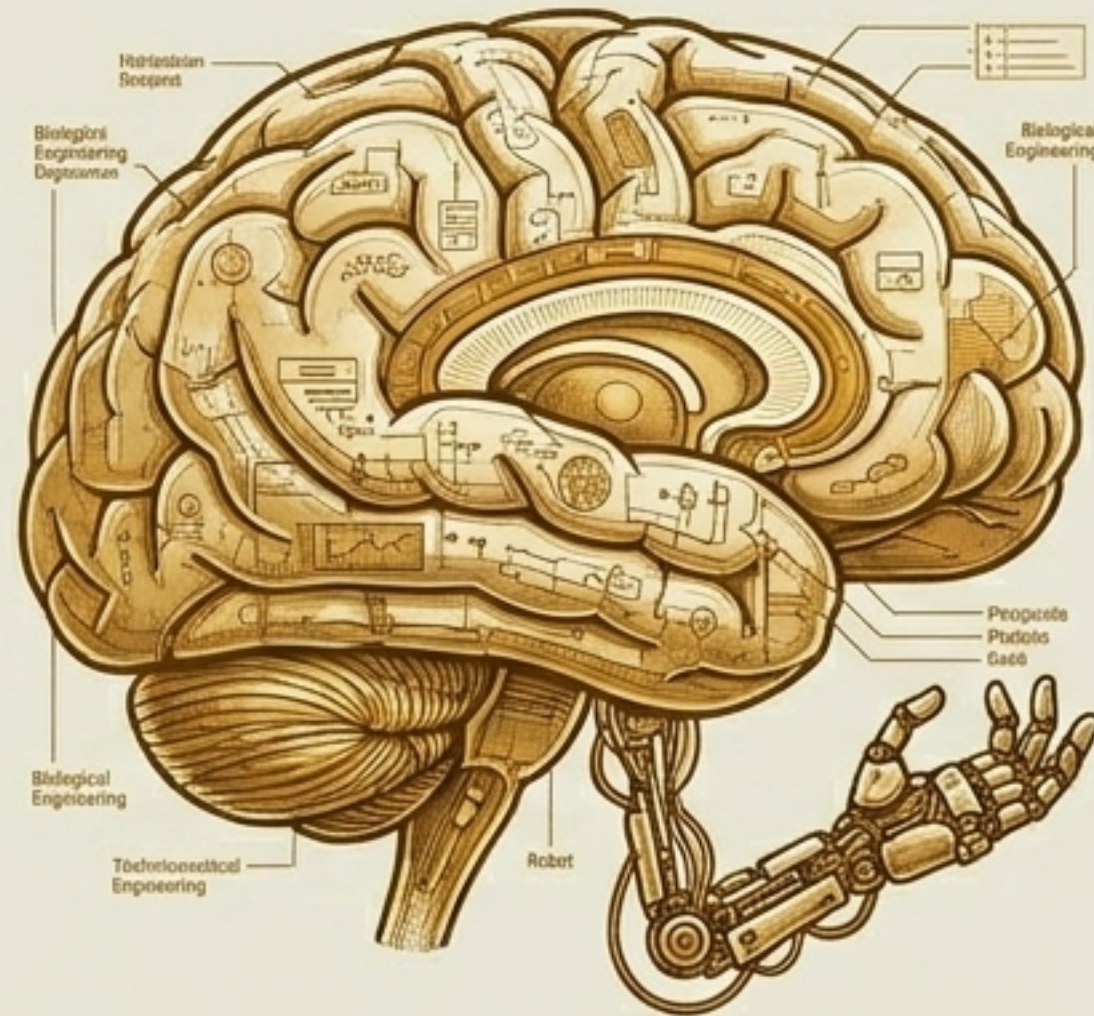
THE ERROR SIGNAL: LEARNING WITHOUT LANGUAGE



If the imagination matches reality, the system stays quiet. If they misalign, a continuous error signal updates the physical world model. This is how the 2022 'Daydreamer' AI learned to walk in just one hour from raw real-world experience—by predicting the next physical states.

THE TWO AI FACTIONS MEETING IN THE MIDDLE

For years, AI research has been split. But these two opposing approaches are finally converging into a single, unified architecture.

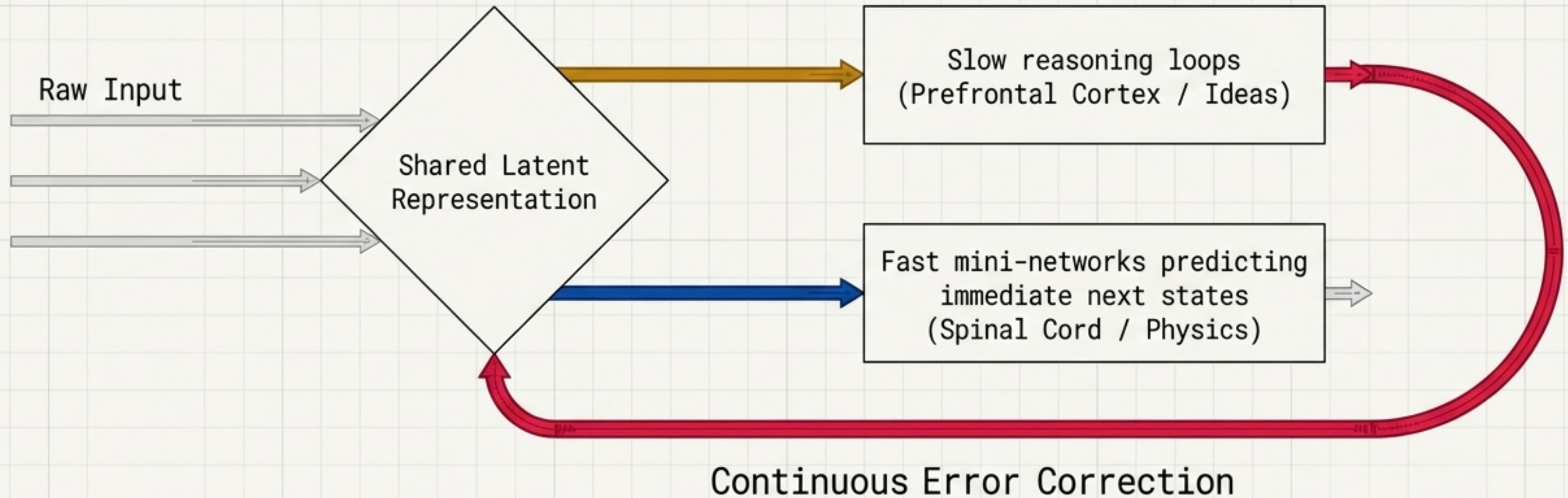


Language-First Paradigm
Massive LLMs emitting actions
as an afterthought.



Action-First Paradigm
Grounded physical world models
layering intelligence on top.

THE UNIFIED BLUEPRINT FOR PHYSICAL INTELLIGENCE



Fast, unconscious physics prediction over the short term. Slow, conscious idea prediction over the long term. Both stemming from the same sensory root, continuously perfected by real-world error signals.

The Revolution is Mass Produced.



The hardware barrier to entry has collapsed. The software divide between language and action has closed.

The future of general purpose robotics isn't locked in a billion-dollar lab—it's arriving as a \$100 kit.

We are launching a small-batch Alpha test to build, train, and refine the next iteration.

[Join the Alpha Test]